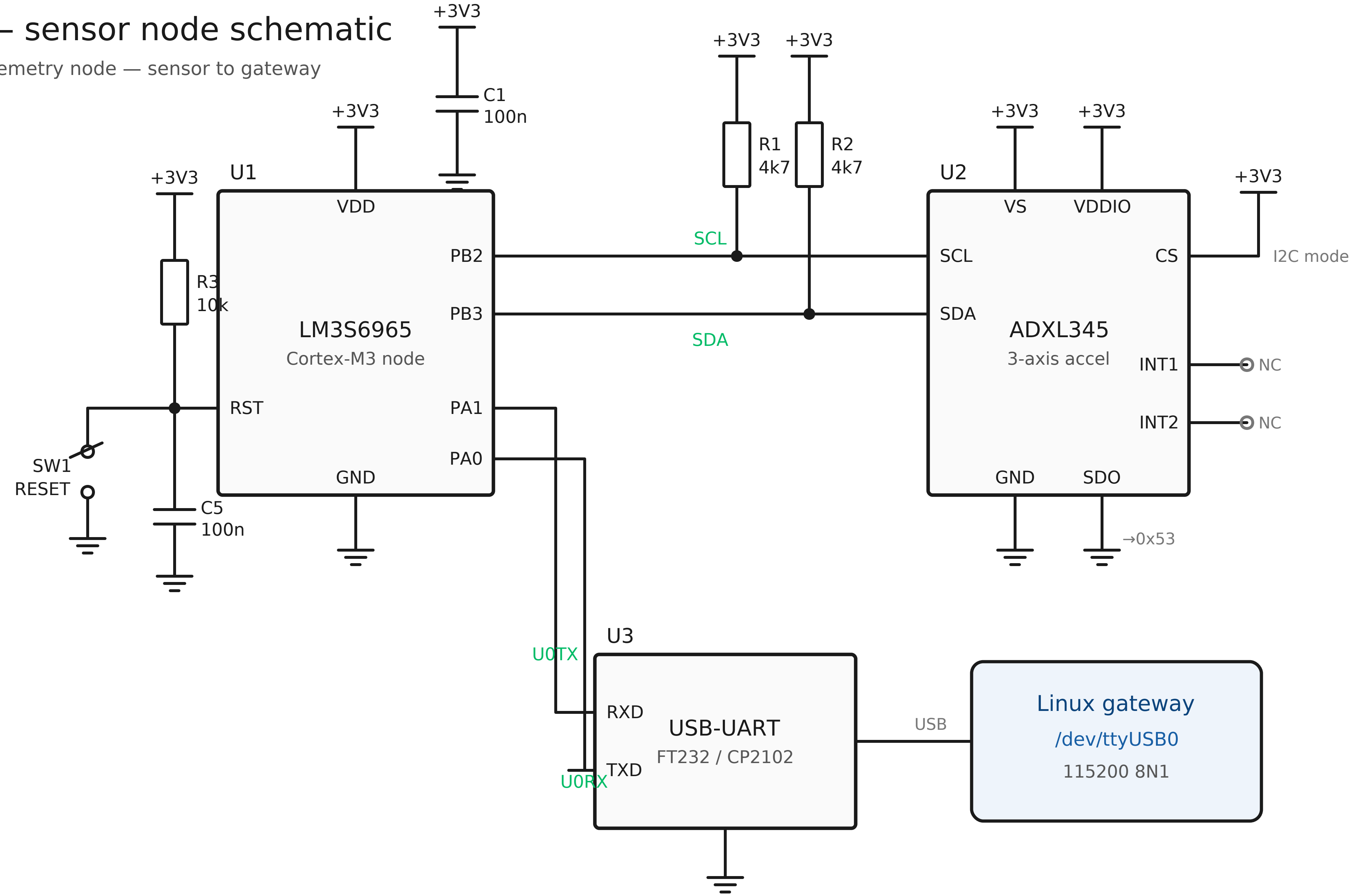
SHARAK — sensor node schematic

IIoT vibration-telemetry node — sensor to gateway

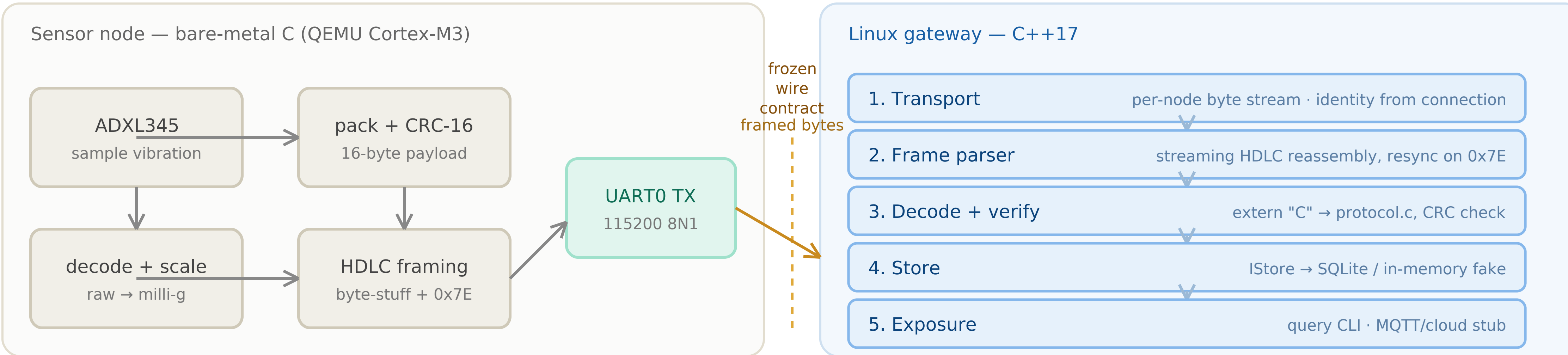


Notes

- Bus: I2C0 single-master @ 100 kHz — PB2=SCL, PB3=SDA, both pulled to +3V3 (open-drain). ADXL345 CS→+3V3 selects I2C; SDO→GND sets 7-bit address 0x53.
- Link: UART0 @ 115200 8N1 — PA1=TX→bridge RXD, PA0=RX→bridge TXD. Off-board USB-serial bridge presents /dev/ttyUSB0 to the gateway host.
- Clock: runs from internal oscillator (PLL bypassed, ~12 MHz) per firmware — no external crystal required. Power: single +3V3 rail; 100 nF decoupling at each supply pin (C1, +ADXL VS/VDDIO).
- Emulation: under QEMU lm3s6965evb the UART link is a TCP socket (127.0.0.1:5555) and I2C is not emulated — the sim backend is injected in its place (see docs/architecture.md §2).

SHARAK — system stages (end to end)

One vertical slice: sensor → node firmware → frozen wire → Linux gateway → storage & exposure



exercised against a fleet of simulated nodes that encode through the same protocol.c

Layer status

- Protocol core · done
- Node firmware · done
- CI (tests + builds) · done
- Gateway (C++17) · in development
- Linux driver · planned
- MQTT + live view · planned
- FreeRTOS / STM32 port · planned
- On-device anomaly detect · planned

What flows on the wire (frozen contract)

payload[16] = ver(1) | type(1) | seq(2,LE) | x_mg(4,LE) | y_mg(4,LE) | z_mg(4,LE)

frame = 0x7E stuffed(payload + CRC16-CCITT [big-endian]) 0x7E (≤ 38 bytes worst case)

identity is NOT on the wire — the gateway derives node ID from the transport (which connection a frame arrived on)